| Reg. No | | | | | | | | | | |
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**St. Joseph's College of Arts & Science (Autonomous)**
**St. Joseph's College Road, Cuddalore – 607001**
**B.Sc Degree Examination – APRIL 2021**

**MT305 – DIFFERENTIAL EQUATIONS**

**Time : 3 hrs** **Max Marks :75**

**SECTION – A (5X2=10)**

Answer **ALL** Questions

1. Solve $p^2 - 5p + 6 = 0$.
2. Solve $\frac{d^2y}{dx^2} + \frac{dy}{dx} - 2y = 0.$
3. Solve $(y+z)\,dx + (z+x)\,dy + (x+y)\,dz = 0.$
4. Solve $p + q = \sin x + \sin y.$
5. Find $L\left[\sin^2 2t\right]$.

**SECTION – B (3X5=15)**

Answer any **THREE** Questions

6. Solve $\frac{dy}{dx} - \frac{dx}{dy} = \frac{x}{y} - \frac{y}{x}.$
7. Solve $x^2 \frac{d^2y}{dx^2} - 3x\frac{dy}{dx} = x + 1.$
8. Solve $\frac{dx}{y - xz} = \frac{dy}{yz + x} = \frac{dz}{x^2 + y^2}.$
9. Solve $p^3 + q^3 = 8z.$
10. Find $L^{-1}\left[\frac{7s-1}{(s+1)(s+2)(s+3)}\right].$

**SECTION – C (5X10=50)**

Answer **ALL** Questions

11. a) Solve $p^2 + 2yp\cot x - y^2 = 0.$

**(or)**

b) Solve $y = 3px + 6p^2y^2$.

12. a) Solve $x^2 \frac{d^2y}{dx^2} - 2x\frac{dy}{dx} - 4y = x^4$.

**(or)**

b) Solve $\frac{d^2y}{dx^2} + y = \cos ecx$.

13. a) Solve $(mz - ny)dx + (nx - lz)dy + (ly - mx)dz = 0$.

**(or)**

b) Solve $\left(ye^x + e^z\right)dx + \left(ze^y + e^x\right)dy + \left(e^y - ye^x - ze^y\right)dz = 0$.

14. a) Solve $Z^2 = xy\ pq$.

**(or)**

b) Solve $x^2p^2 + y^2q^2 = z$.

15. a) Solve $\frac{d^2y}{dt^2} - \frac{dy}{dt} - 2y = 0$ given $y(0) = -2, y^1(0) = 5$.

**(or)**

b) Solve $\frac{d^2y}{dx^2} + 5\frac{dy}{dx} + 6y = e^{-x}$ given $y = 0, \frac{dy}{dx} = 1$, when $x = 0$ using Laplace transform.

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